

Glider Design Report

Aerodynamics – Fall 2024

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Introduction

Our team was assigned to design a glider that will maximize glide distance and maximize payload to glider weight ratio. This report will walk through the glider design process and compare the estimated performance to the actual flight testing. Drawings, diagrams, and calculations will all be used to show how the glider came to be. Furthermore, a bill of materials will be included detailing the location of purchase and cost of the glider parts.

Design Process

To begin the design process, the glider was decided to have a conventional configuration. After establishing the basic arrangement of the glider, several key parameter values were chosen. Using the max wing planform area of 480in^2 , a wingspan and chord were picked to be 60 inches and 8 inches respectively. From here, the tail dimensions were chosen to equate to half of the wing's planform area (240in^2). The chord was designed at 6 inches, making the span 40 inches. The vertical tail was designed at a chord of 6 inches for simplicity of construction.

For the wing and horizontal tail airfoil, a NACA4412 airfoil was used due to its high positive coefficient of lift at low angles of attack. The NACA0012 airfoil was used for the vertical tail because it is uncambered and symmetrical.

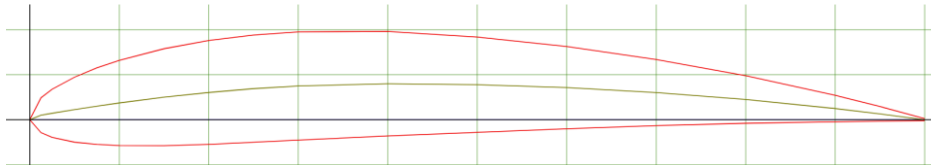


Figure 1: NACA4412 Wing Profile (Used for Main Wing and Horizontal Tail)-+3

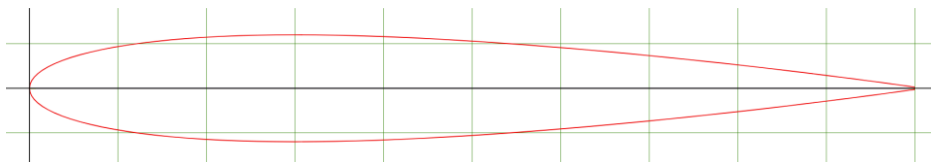


Figure 2: NACA0012 Wing Profile (Used for Vertical Tail)

The payload compartment was designed at the front of the glider to improve stability by keeping the center of gravity well in front of the neutral point position.

Stability of flight was the main concern in this design. The team targeted a 65%-85% margin of stability, so many design characteristics were calculated based on this parameter. The design utilized a wing and tail dihedral at small angles to improve straight flight while not compromising lift. 3D modeled wing seats were designed to make sure the wing and tail components sat at the correct angles and were aligned properly. Overall, the design was very simple to avoid human error in construction.

Here is a summary of our chosen design parameters:

Chosen Design Parameters		
Wing Chord	c_w	8"
Wing Span	b_w	60"
Horizontal Tail Chord	c_t	6"
Horizontal Tail Span	b_t	40"
Vertical Tail Chord	c_{vt}	6"
Vertical Tail Span	b_{vt}	8"
Tail Incident Angle	i_t	-1 deg

Our Team Made a Bad Error

After conducting the final flight testing, we saw major issues with stability and flight characteristics when the glider was being launched from a full height of 22 feet. The drawing in Figure 3 shows the flight characteristics of the glider during official flight testing.

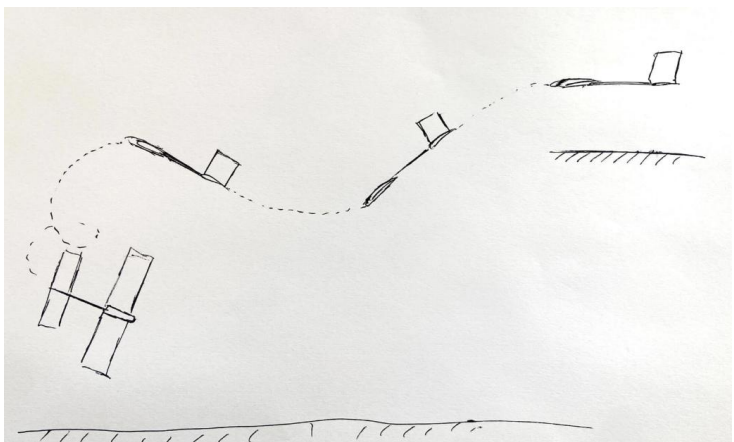


Figure 3: Diagram of issues during official flight testing.

As can be seen, once the glider began to achieve lift, it would pitch upwards, causing a decrease in glide speed and increase in angle of attack. This would stall the glider and

cause it to fall from the sky. We will talk about the specifics in the flight analysis later in the report—however, we were completely stumped as to why our glider was having issues. It wasn't until after flight testing that we went back through all of our calculations and realized that we had made a math error. We aren't entirely sure how we made the mistake, but the CG of the glider ended up about 6.5 inches *behind* the neutral point. Not great. The highest likelihood is that during initial stability calculations, we accidentally swapped the neutral point and center of gravity locations.

$$\text{Margin of Stability, } MOS = \frac{l_{wn} - l_{CG}}{c_w}$$

$$\text{Our Mistake: } MOS_{\text{wrong}} = \frac{l_{CG} - l_{wn}}{c_w}$$

Because the result ended up positive, the error went by unnoticed, and it was constructed. The target stability margin was intended to be in the range of 50-70% to ensure that manufacturing errors wouldn't induce stability issues. However, due to the math error, the glider margin of stability worked out to be around 70%, but in the wrong direction—it was unstable. Therefore, all of the diagrams and calculations will demonstrate the glider in this originally designed configuration even though it is unstable.

Final Glider Design

Drawings

After the initial design phase, the glider was modeled in SolidWorks. This enabled the team to implement design for manufacturing practices and to analyze the position of components on the glider. The following drawings are of the Right, Front, Top, and a Section View. They demonstrate the base dimensions of the glider, including the location of aerodynamic centers, the center of gravity, neutral point, and payload location.

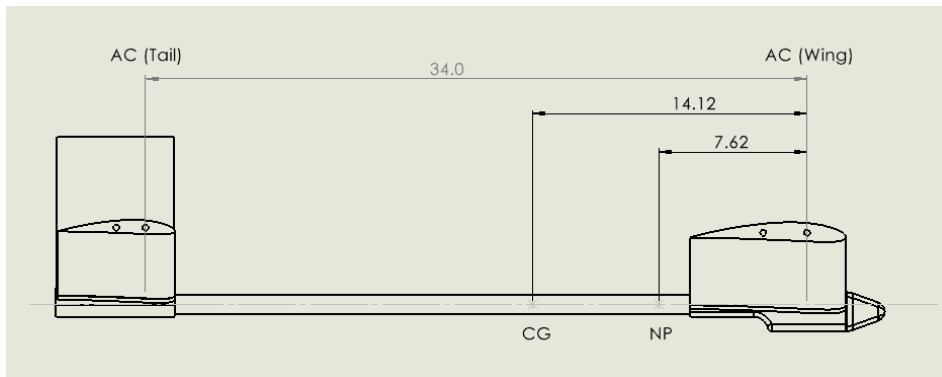


Figure 4: Glider Right Side View Drawing (Includes Neutral Point, Center of Gravity, and Aerodynamic Centers)

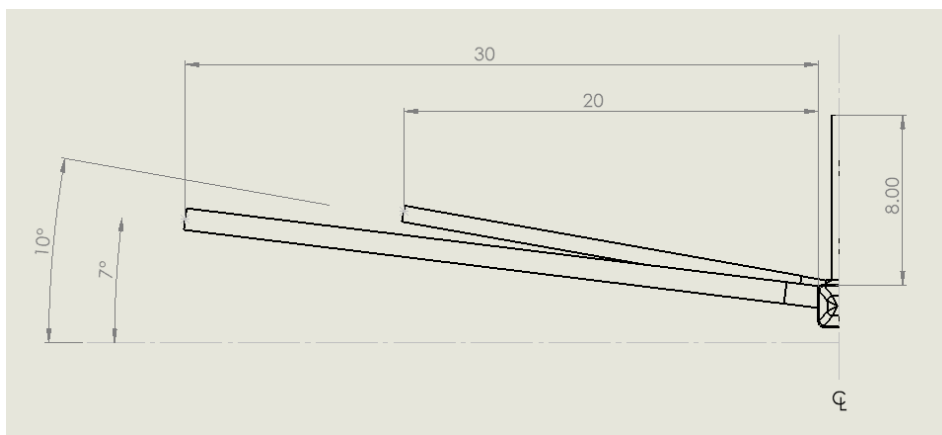


Figure 5: Glider Front Side View Drawing (Includes Half Wing and Tail Spans, Dihedral Angles of Wing and Tail)

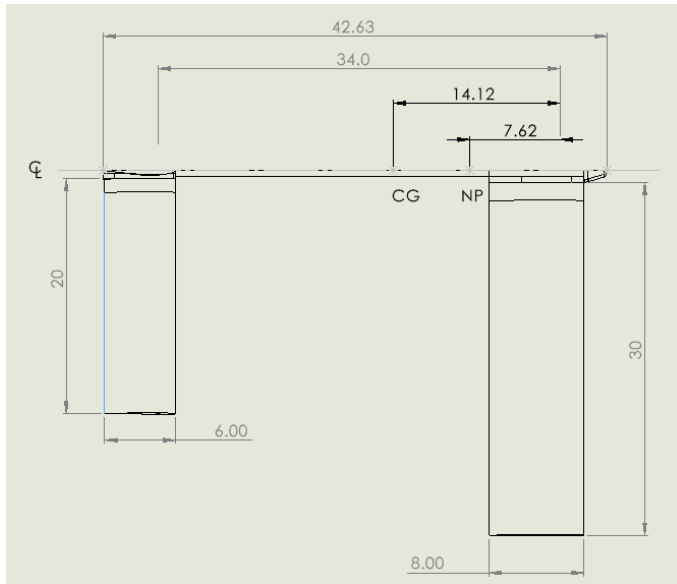


Figure 6: Glider Top View (Includes Half Spans, Chord Lengths, and CG/NP Locations)

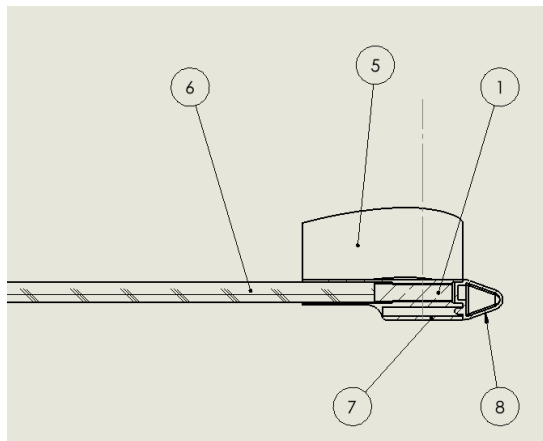


Figure 7: Section View of Fuselage and Payload Area.

In Figure 7, a section view of the fuselage and payload area is shown. The following components are included in the view: (1) Payload, (5) Main Wing, (6) Fuselage Dowel, (7) Main Wing Mounting Body, and (8) Nose Cone. Not shown are the tail components. Figure 8 shows the location and capacity of the payload in an isometric image of the glider.

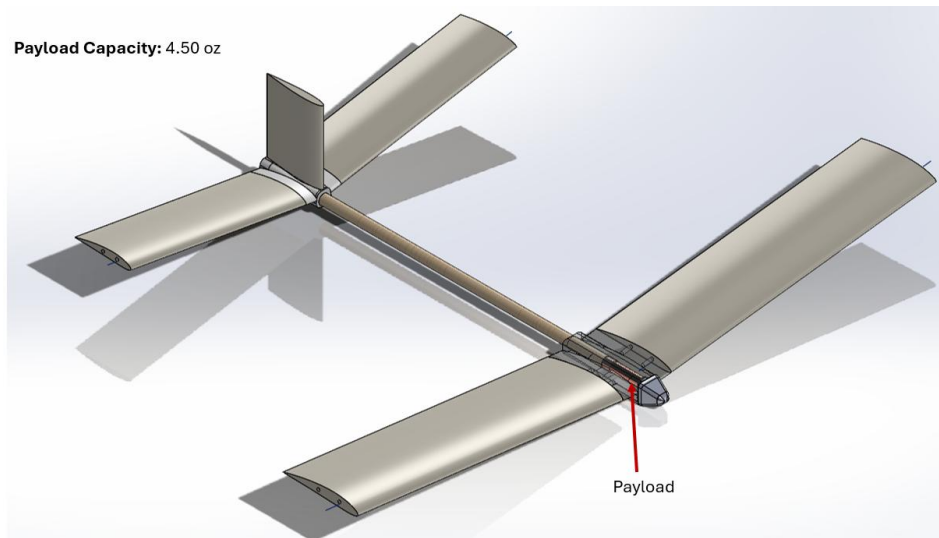


Figure 8: Isometric View of Glider with listed payload capacity and location of payload.

Construction

Bill of Materials

After the glider design process was completed, a bill of materials (BOM) for components was constructed. The BOM reflects each component, and the quantities used in the construction of the glider. It is shown in Figure 9. The details on individual components are described later in this section.

Bill of Materials			
Component:	Quantity:	Unit Weight (oz):	Net Weight (oz):
Main Wing Dowel	4	0.66	2.64
Horizontal Tail Dowel	4	0.52	2.08
Main Wing Element	6	0.63	3.78
Horizontal Tail Element	4	0.35	1.4
Vertical Stabilizer Element	1	0.41	0.41
Main Wing Body Mount	1	7.38	7.38
Horizontal Tail Body Mount	1	3.70	3.7
Fuselage Dowel	1	10.55	10.55
Nose Cone	1	0.93	0.93
10-24 x 2.5" Screw	5	0.54	2.7
10-24 Anti-Vibration Locknut	3	0.06	0.18
Hot Glue	1	7.15	7.15
Total Glider Weight (oz):			42.9

Figure 9: Bill of Materials for the Glider Construction

Expenditures

The cost of the glider was broken down into two counterparts—raw material cost and net glider cost. The cost of the raw materials includes the foam, dowels, PLA 3D printing filament, and hot glue if the team was to purchase the materials in bulk. This cost breakdown can be seen in Figure 10. The total cost to purchase all glider materials in bulk would be approximately \$102.68. Since many of the materials were already available from the university, it did not accurately reflect the total cost to build the glider. Therefore, an itemized list and cost-by-weight of the glider components was used to evaluate the actual glider cost. The final list of expenditures was created based on cost-by-weight of components used, resulting in a net glider cost of \$31.71 (see Figure 11).

Bulk Materials						
Item	Quantity	Cost per Unit	Total Cost	Bulk Weight (oz)	Cost/Weight (\$/oz)	Purchase Location
2"x4"x8' Foamular NGX Insulation Foam	1	\$ 54.98	\$ 54.98	120	\$ 0.46	Home Depot
5/16"x48" Hardwood Round Dowel	4	\$ 1.07	\$ 4.28	4.48	\$ 0.96	Home Depot
1"x36" Hardwood Round Dowel	1	\$ 8.44	\$ 8.44	10.55	\$ 0.80	Home Depot
50pk Hot Glue Sticks	1	\$ 7.99	\$ 7.99	16.8	\$ 0.48	Harbor Freight
Prusa PLA 3D Printing Filament	1	\$ 26.99	\$ 26.99	35.27	\$ 0.77	Prusa
Bulk Material Cost:						\$ 102.68

Figure 10: Bulk Material Expenditures

Glider Net Cost				
Item	Quantity	Unit Weight:	Material:	Cost:
Main Wing Dowel	4	0.66	5/16" Dowel	\$ 2.52
Horizontal Tail Dowel	4	0.52	5/16" Dowel	\$ 1.99
Main Wing Element	6	0.63	Insulation Foam	\$ 1.73
Horizontal Tail Element	4	0.35	Insulation Foam	\$ 0.64
Vertical Stabilizer Element	1	0.41	Insulation Foam	\$ 0.19
Main Wing Body Mount	1	7.38	PLA	\$ 5.65
Horizontal Tail Body Mount	1	3.7	PLA	\$ 2.83
Fuselage Dowel	1	10.55	1" Hardwood Dowel	\$ 8.44
Nose Cone	1	0.93	PLA	\$ 0.71
10-24 x 2.5" Screw	5	0.54	---	\$ 1.74
10-24 Anti-Vibration Locknut	3	0.06	---	\$ 1.87
Hot Glue	1	7.15	---	\$ 3.40
Total Glider Cost:				\$ 31.71

Figure 11: Net Glider Cost Breakdown Table

Fuselage

For the fuselage, a 36" long, 1" diameter oak dowel was used. The dowel fit into the wing and tail seat and served as the connection point between the wing and the tail.

Wing and Horizontal Tail

The wing was constructed into two different sections at $\frac{1}{2}$ of the span. Each half span was cut into thirds, and two 5/16" diameter wooden dowels were used to support and connect the three sections together. A 1" length was left at the end of the dowels that press fit into the wing seat holes.

To cut the sections of the wings and horizontal tail, thin cutting templates (1/4") of the NACA4412 at 8" and 6" chord respectively were printed from PLA. A layer of masking tape was applied to the foam and the cutting template, then spray adhesive was used to attach the masked components together. This enabled the team to reuse the templates without getting spray adhesive on the foam. The spray adhesive melts the foam if it is used directly on the material. The foam was then fastened to the table, and a hot wire used to cut 9.5" long airfoil sections for both 8"- and 6"-chord airfoils. This step ended with 6 individual profile sections for the wing and 4 individual profile sections for the tail.

After the airfoil profiles were cut, a second template with trough features was then attached to the airfoil. These features were 0.25" wide, ~0.6" deep troughs that would be used as channels for the dowels.

The inserts were then cut out of the bottom of the wing and tail using these newly printed profiles. A picture of these profiles can be found below in Figure 12. These second templates could have been used for the whole process, however, we did not develop the idea for this style of template until all of the airfoils had already been cut out. Initially, the plan was to use a fixture with a hot metal rod to create holes through the foam, but we developed this as a much better method.



Figure 12: Slotting template for the airfoil profiles.

The template created undersized slots that enabled a press fit for the 5/16" dowels. Finally, the dowels were hot glued into the inserts to finish the wing and horizontal tail. Masking tape was used to cover the imperfections in the surface finish.

Vertical Tail

The vertical tail was cut into one piece using the wire cutter setup. 3D printed templates were also used to provide the airfoil profile. A symmetrical NACA0012 airfoil profile was used. Holes were drilled in the airfoil for locating, and dowels inserted through the main mounting components fixtured the vertical stabilizer in place—similar to the main wing and tail. Hot glue was used to secure everything in place.

Wing and Tail Seat

The wing and tail seats were 3D printed in the ME department's Maker Space lab using PLA at 10% infill. Because the wing seat was printed in 2 sections, the pieces were clamped together with a screw and nut.

The wing seat also served as the payload compartment with a 0.875" diameter, 3" deep hole for the roll of quarters to fit into. A nose cone was designed to be a removable cover

for the payload compartment. This compact and aerodynamic design of the payload compartment minimized profile drag and kept construction very simple.

Assembly

The assembly of the glider was relatively straight forward. Firstly, the individual wing and tail section had to be attached. This was achieved by simply gluing the wooden dowels into the cutout slots on the bottom of the wing and tail. The 1" dowel overhang was then inserted into the 3D printed wing and tail housing, completing the wing and horizontal tail assembly. Finally, the vertical tail was glued into its slot in the tail housing. The wing and both tails were reinforced with tape to add some extra reinforcement. The finished glider can be seen below.

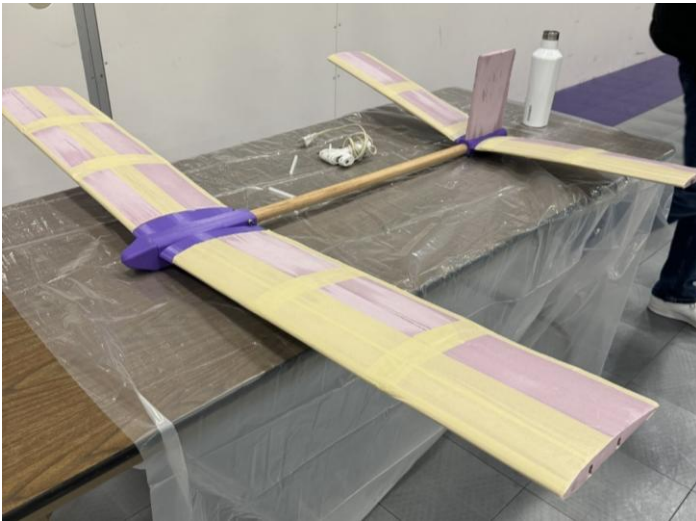


Figure 13: Completed glider prior to final flight testing at the Recreation Center.

Predicted Glider Flight Performance

In this section, we address different design aspects of the glider and predict their performance prior to flight testing.

C.G. Location

The first design parameter that we predicted was the glider center of gravity (CG). Using SolidWorks, the CG for each component in the glider construction was located. A reference plane was placed on the very front surface of the glider's nose cone, and the distance from the front plane to the component CG was measured via SolidWorks' evaluation tool. Due to the varying densities of materials from suppliers, we weren't able to accurately predict the masses of our components before we obtained the materials in hand. Therefore, we elected to measure the mass of each individual component and assembly after it was manufactured.

The results are listed below:

Glider Center of Gravity Location			
All distances are measured from the front of the nose cone.			
Description:	Weight (oz):	Distance (in):	Weight*Distance (oz-in)
Nose Cone	0.93	1.46	1.36
Main Wing Mount	7.38	5.44	40.15
Right Wing Assembly	4.57	5.37	24.54
Left Wing Assembly	4.66	5.37	25.02
Fuselage Dowel	10.55	24.375	257.16
Horizontal Tail Mount	4.78	39.33	188.00
Right Tail Assembly	2.5	39.06	97.65
Left Tail Assembly	2.5	39.06	97.65
Vertical Stabilizer	0.41	39.1	16.03
Payload	4.5	4.4375	19.97

Figure 14: Compiled Center of Gravity Data

To calculate the overall CG location of the glider, we used the following equation, where X_{CG} is the location of the CG with respect to the front plane, W_i is the component weight, and x_i is the distance of the component CG from the front plane.

$$\text{Center of Gravity Equation: } x_{cg} = \frac{\sum W_i x_i}{\sum W} = 17.89" \quad (\text{from glider leading edge})$$

Using this location of the center of gravity relative to the front of the glider, we can find the distance from the wing's aerodynamic center (AC) to the center of gravity location. By adding the nosecone's length of 2" and the wing's quarter chord of 2", the wing's AC

location is 4" behind the front of the glider. Using these values, the location of the center of gravity relative to the wing's AC is shown below:

$$CG \text{ Location from the Wing AC, } l_w = x_{cg} - 4" = 13.89"$$

Attitude Angle

To calculate the attitude angle of the glider, the equation for the moment coefficient about the center of gravity is set equal to zero because there should be no pitching moment in steady glide.

$$C_{mcg} = \frac{l_w}{c_w} C_{Lw} - V_{Ht} C_{Lt} = 0$$

$$X = l_w + l_t, \quad l_t = X - l_w$$

$$V_{Ht} = \frac{l_t S_t}{c_w S_w} = \frac{(20.11in)(240in^2)}{(8in)(480in^2)} = 1.257$$

The slope of the section lift coefficient for the wing and tail is found from the C_l vs. α plot for NACA4412 by taking the slope of the linear range (-5deg to 10deg).

$$a_w = \frac{dC_{lw}}{d\alpha} = 0.09 \text{ deg}^{-1}$$

$$a_t = \frac{dC_{lt}}{d\alpha} = 0.09 \text{ deg}^{-1}$$

Expressions for total lift coefficients for the wing and tail are shown below. Because a T-tail configuration is used, downwash effect is negligible. The incident angle of the tail is found to be about 1 degree.

$$C_{Lw} = a_w \alpha = (0.09)\alpha$$

$$\varepsilon(\alpha) = 0$$

$$C_{Lt} = a_t(\alpha - i_t - \varepsilon(\alpha)) = 0.09(\alpha - 1 - 0) = 0.09(\alpha - 1)$$

Substituting the lift coefficient expressions back into the moment equation and solving for α :

$$\alpha = \frac{-V_{Ht} a_t i_t}{\frac{l_w}{c_w} a_w - V_{Ht} a_t} = \frac{-1.257(0.09)(1)}{\frac{13.89in}{8in}(0.09) - 1.257(0.09)} = 2.6 \text{ deg}$$

Glide Speed

The following equation for dynamic pressure will be used to start the glide speed calculations:

$$q = \frac{1}{2} \rho v_g^2 = \frac{L_T}{C_{LT} S_T}$$

Solving for v_g ...

$$v_g = \sqrt{\frac{2L_T}{\rho S_T C_{LT}}}$$

Firstly, we know...

$$\rho = 0.072 \frac{\text{lbm}}{\text{ft}^3}$$

For lift, we know...

$$L_T = W \cos(\alpha_g)$$

So we will assume a small angle and say...

$$L_T \approx W = 42.9 \text{ ozf}$$

For the wing and tail lift coefficients, the section lift coefficients the airfoil will be used to calculate...

$$C_{Lw} = \frac{c_l}{1 + \frac{2}{AR_w}}$$

$$C_{Lt} = \frac{c_l}{1 + \frac{2}{AR_t}}$$

Commented [AM1]: Question for myself for clarification, is this t for total or t for tail??

The section lift coefficients will be found using an assumed 5 degree angle of attack and the below airfoil section data...

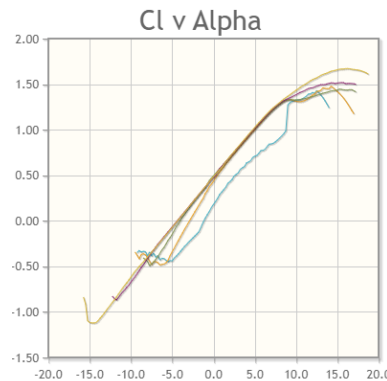


Figure 15: Section Lift Coefficient vs. Angle of Attack, NACA4412 Airfoil

Using all of this...

$$C_{Lw} = \frac{1.1}{1 + \frac{2}{\frac{60}{8}}} = 0.868$$

$$C_{Lt} = \frac{1.1}{1 + \frac{2}{\frac{38}{6}}} = 0.836$$

Finally, solving for glide speed...

$$v_g = \sqrt{\frac{2 * 42.9 \text{ ozf} * \frac{1 \text{ lbf}}{16 \text{ ozf}}}{0.072 \frac{\text{lbm}}{\text{ft}^3} * \frac{1 \text{ lbf}}{32.2 \text{ lbm} \frac{\text{ft}}{\text{s}^2}} * \frac{1 \text{ ft}^2}{144 \text{ in}^2} * (228 \text{ in}^2 + 480 \text{ in}^2) * (0.868 + 0.836)}} = 16.9 \frac{\text{ft}}{\text{s}}$$

Glide Angle

To start, a force balance yields...

$$W \sin(\alpha_g) = D_T$$

Since the weight is already known, solving for D_T will allow the glide angle to be solved for.

Total drag can be described as...

$$D_T = D_i + D_p = q \sum C_D S$$

The coefficient of drag for both the wing and tail needs to be found to find total drag. These can be found through these equations...

$$C_{Dw} = C_{dw} + \frac{C_{Lw}^2}{\pi(AR_w)}$$

$$C_{Dt} = C_{dt} + \frac{C_{Lt}^2}{\pi(AR_t)}$$

The section drag coefficients can be pulled off the graph below using a 5 degree alpha...

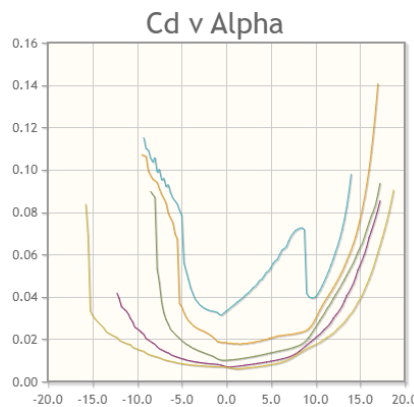


Figure 16: Section Drag Coefficient vs. Angle of Attack, NACA4412 Airfoil

Using the lift coefficients and aspect ratios from previous parts the drag coefficients can be found...

$$C_{Dw} = 0.02 + \frac{0.868^2}{\pi(7.5)} = 0.0520$$

$$C_{Dt} = 0.02 + \frac{0.836^2}{\pi(6.33)} = 0.0551$$

Taking these coefficients and turning them into forces...

$$D_w = qSC_{Dw} = 0.5 * \left(16.9 \frac{ft}{s}\right)^2 * 0.072 \frac{lbm}{ft^3} * \left(\frac{1 lbf}{32.2 lbm \frac{ft}{s^2}}\right) * 3.333 ft^2 * 0.0520 * \frac{16 ozf}{1 lbf} = 0.885 ozf$$

$$D_t = qSC_{Dt} = 0.5 * \left(16.9 \frac{ft}{s}\right)^2 * 0.072 \frac{lbm}{ft^3} * \left(\frac{1 lbf}{32.2 lbm \frac{ft}{s^2}}\right) * 1.583 ft^2 * 0.0551 * \frac{16 ozf}{1 lbf} = 0.446 ozf$$

Next the parasite drag will be estimated...

$$D_{fuselage} = qA_{xsect}C_D$$

Table 11-1 from Cengel and Cimbala's Fluid Mechanics: Fundamentals and Applications will be used to approximate the fuselage drag. The nose can be best estimated to be the bullet shape seen below, labeled as a round front edge...

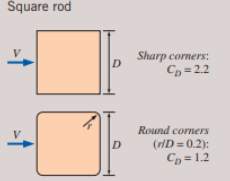
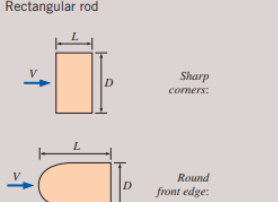
TABLE 11-1																									
Drag coefficients C_D of various two-dimensional bodies for $Re > 10^4$ based on the frontal area $A = bD$, where b is the length in direction normal to the page (for use in the drag force relation $F_D = C_D A \rho V^2 / 2$ where V is the upstream velocity)																									
Square rod  Sharp corners: $C_D = 2.2$ Round corners ($r/D = 0.2$): $C_D = 1.2$	Rectangular rod  Sharp corners: <table border="1"> <thead> <tr> <th>L/D</th><th>C_D</th></tr> </thead> <tbody> <tr><td>0.0*</td><td>1.9</td></tr> <tr><td>0.1</td><td>1.9</td></tr> <tr><td>0.5</td><td>2.5</td></tr> <tr><td>1.0</td><td>2.2</td></tr> <tr><td>2.0</td><td>1.7</td></tr> <tr><td>3.0</td><td>1.3</td></tr> </tbody> </table> * Corresponds to thin plate Round front edge: <table border="1"> <thead> <tr> <th>L/D</th><th>C_D</th></tr> </thead> <tbody> <tr><td>0.5</td><td>1.2</td></tr> <tr><td>1.0</td><td>0.9</td></tr> <tr><td>2.0</td><td>0.7</td></tr> <tr><td>4.0</td><td>0.7</td></tr> </tbody> </table>	L/D	C_D	0.0*	1.9	0.1	1.9	0.5	2.5	1.0	2.2	2.0	1.7	3.0	1.3	L/D	C_D	0.5	1.2	1.0	0.9	2.0	0.7	4.0	0.7
L/D	C_D																								
0.0*	1.9																								
0.1	1.9																								
0.5	2.5																								
1.0	2.2																								
2.0	1.7																								
3.0	1.3																								
L/D	C_D																								
0.5	1.2																								
1.0	0.9																								
2.0	0.7																								
4.0	0.7																								

Figure 17: Coefficient of Drag for various shapes.

With the L/D ratio being at least 2.0, the C_D can be read as 0.7. Using this information...

$$D_{fuselage} = qA_{xsec}C_D$$

$$= 0.5 * 0.072 \frac{lbm}{ft^3} * 16.9 \frac{ft^2}{s} * \left(\frac{1 lbf}{32.2 lbm \frac{ft}{s^2}}\right) * \left(\frac{1 ft^2}{144 in^2}\right) * \frac{16 ozf}{1 lbf} * 2in * 2in * 0.7$$

$$= 0.0993 ozf$$

Now to add up all the drag forces...

$$D_{total} = D_w + D_t + D_{fuselage} = 0.794 \text{ ozf} + 0.399 \text{ ozf} + 0.0993 \text{ ozf} = 1.29 \text{ ozf}$$

And find the glide angle...

$$\alpha_g = \sin^{-1} \left(\frac{D_{total}}{W} \right) = \sin^{-1} \left(\frac{1.29 \text{ ozf}}{42.9 \text{ ozf}} \right) = 1.73^\circ$$

Neutral Point Location

To find the neutral point, the below equation will be used...

$$l_{wn} = X \frac{S_t}{S_t + S_w} = 34in \frac{240in^2}{240in^2 + 480in^2} = 8.38in$$

Where X is the distance between the quarter chord positions of the wing and tail, and S_t and S_w are the tail and wing planform area respectively. l_{wn} is the distance from the wing aerodynamic center to the neutral point.

Assessment of Stability

The Margin of Stability (MOS) equation will be used to assess stability...

$$MOS = \frac{l_{wn} - l_w}{c_w} * 100$$

Where $l_{wn} - l_w$ is the distance between the neutral point and the center of gravity, and c_w is the chord of the wing.

Solving...

$$MOS = \frac{l_{wn} - l_w}{c_w} * 100 = \frac{8.38 - 13.89 \text{ in}}{8 \text{ in}} * 100 = -69\%$$

Recalling the “Our Team Made a Bad Mistake Section”, this is what we were discussing. Initially, our team believed that the glider possessed an MOS of 69%, however, that was not true. Due to the math error, we improperly analyzed this characteristic, and the true stability of the glider is -69%. It is unstable. This provides the reasoning for the flight testing characteristics that we discuss later in the report.

Quantitative Comparison

Official Testing Results

Flight Testing	Payload Weight (oz)	Payload to Empty Weight Ratio	Glide Characteristics (Comments)	Vertical Elevation VE (ft)	Longitudinal Distance D (ft)	Lateral Distance L (ft)	Flight Duration (s)
Trial #1	4.50	0.117	flipped over, no stable glide, either longitudinally unstable, or wrong tail incidence	22	41.5	2.0	2.10
Trial #2	4.50	0.117	more gentle throw, but glide never developed, unstable or wrong tail incidence	22	34.7	2.5	2.52
Trial #3	4.50	0.117	better initial "glide", but never developed clear stable glide, may be close to neutral stability	22	42.0	3.5	2.57
Trial #4	4.50	0.117	improved a tad, but appears to stall or have unstable moment--suggest checking stability	22	39.5	11.0	3.05
Trial #5	4.50	0.117	after launch, nose dives this time, crashes to floor	22	29.5	3.3	1.60

****note: Trial #5 was done with an added weight attached in place of the designed nosecone. This trial was done to simply determine if more weight at the front of the glider could help solve any issues, therefore the data is not included in any calculations or analysis of our glider****

Figure 18 - Compiled Official Test Results

Actual Performance Parameters and Results

Parameters		
Span, Wing	b_w	60 in
Chord, Wing	c_w	8 in
Planform Area, Wing	S_w	480 in ²
Span, Tail	b_t	40 in
Chord, Tail	c_t	6 in
Planform Area, Tail	S_t	240 in ²
Empty Weight	W_0	38.4 ozf
Weight with Payload	W	42.9 ozf
Air Density	ρ	0.072 lbm/ft ³

Figure 19: Glider performance parameters.

Trial	Glide Distance G (ft)	Glide Speed Vg (ft/s)	Glide Angle (deg)
1	47.01	22.39	27.90
2	41.16	16.33	32.31
3	47.54	18.50	27.56
4	46.53	15.26	28.22

Figure 20: Glide speed and glide angle results from physical testing.

The glide distance, G, can be found as shown below (for trial 3).

$$G = \sqrt{D^2 + L^2 + h^2} = \sqrt{42 \text{ ft}^2 + 3.5 \text{ ft}^2 + 22 \text{ ft}^2} = 47.54 \text{ ft}$$

Glide Angle

The actual glide angle can be found from the inverse sine of the vertical elevation divided by the glide distance. An example calculation for trial 3 is shown below.

$$\alpha_g = \sin^{-1}\left(\frac{VE}{G}\right) = \sin^{-1}\left(\frac{22 \text{ ft}}{47.54 \text{ ft}}\right) = 27.56 \text{ deg}$$

Glide Speed

The actual glide speed is calculated by taking the total distance the glider travelled and dividing by the glide time. The trial 3 glide speed is calculated below.

$$v_g = \frac{G}{t} = \frac{47.54 \text{ ft}}{2.57 \text{ s}} = 18.50 \frac{\text{ft}}{\text{s}}$$

Lift, Drag, and Coefficients of Lift and Drag

The average glide angle and glide speed were found using the average() function in excel. These results are shown below, along with the other calculated parameters.

Calculated Testing Parameters		
Avg Glide Speed	18.12	ft/s
Avg Glide Angle	29.00	deg
Lift	37.52	ozf
Drag	20.80	ozf
q	0.041	ozf/in ²
C _L	1.277	
C _D	0.708	

Figure 21: Calculated flight characteristics based on flight testing results.

The lift and drag forces were calculated using the average glide angle as follows.

$$L = W * \cos(\alpha_g) = 42.9 \text{ ozf} * \cos(18.12) = 37.52 \text{ ozf}$$

$$D = W * \sin(\alpha_g) = 42.9 \text{ ozf} * \sin(18.12) = 20.80 \text{ ozf}$$

From the calculated lift and drag forces, the coefficients of lift and drag were found using the following equations.

$$q = \frac{1}{2} * \rho * v_g^2 = \frac{1}{2} * 0.072 \frac{lbm}{ft^3} * \left(18.12 \frac{ft}{s}\right)^2 \left(\frac{1 lbf}{32.174 lbm \frac{ft}{s^2}}\right) * \left(\frac{1 ft^2}{144 in^2}\right) * \frac{16 ozf}{1 lbf}$$

$$= 0.0408 \frac{ozf}{in^2}$$

$$C_L = \frac{L}{q * S} = \frac{37.52 ozf}{0.0408 * 720 in^2} = 1.277$$

$$C_D = \frac{D}{q * S} = \frac{20.8 ozf}{0.0408 * 720 in^2} = 0.708$$

Testing Results vs. Predicted Performance

The comparable parameters are summarized below.

Tested vs Predicted Results		
	Predicted	Tested
Glide Angle (deg)	1.79	27.6
Glide Distance (ft)	728	47.5
Glide Speed (ft/s)	16.9	18.5
Lift (ozf)	42.9	37.5
Drag (ozf)	1.29	20.8
Coefficient of Lift	0.852	1.27
Coefficient of Drag	0.02	0.708

Figure 22: Comparison of predicted vs. tested results for the glider.

As shown in the table above, the calculated parameters from our testing data are very different from what we predicted. The biggest discrepancies are the glide angle, glide distance, drag, and coefficient of drag. However, the tested glide angle is not accurate, as our glider never actually achieved steady glide. This issue is discussed more in the conclusion.

Conclusion

In this project, we designed a glider from the ground up with the intent to achieve the furthest glide distance while carrying a payload. We were provided with a set of required parameters—including a launch height of 22 feet, a payload capacity of 4.5 ounces, and the goal to achieve the highest weight-to-carrying capacity possible. A maximum planform area of 480 inches was also specified.

In the initial design process, we chose the NACA 4412 and NACA 0012 airfoils due to their simple profiles, high lift at low angles of attack, and our general familiarity with the airfoil as it was commonly used in class. The team elected to maximize the planform area of the wing, and a span of 60" was selected. With a target Margin of Stability of 50-70%, the rear tail size and aerodynamic center locations were iterated to acquire a Center of Gravity and Margin of Stability within these characteristics.

The glider was then manufactured using a combination of wooden dowels, insulation foam, and 3D prints. As we manufactured the glider, mass and CG positions were evaluated. These were then used to predict the stability, center of gravity, glide speed, and glide angle of the glider.

When flight tested, the team experienced many issues with longitudinal stability. As the glider attempted to reach steady glide, it would be unable to recover from upward pitching of the aircraft. This would cause a stall situation, and we were never able to get the glider to reach steady glide when launched from the official test height. The team attempted to remedy this by implementing trim tabs on the horizontal stabilizer, which did show significant improvements. However, the glider still never reached steady glide with these adjustments.

After the official flight testing, the team realized that we had made a critical math error while predicting our Margin of Stability. Due to this error, the glider was unstable by a margin of 70%, causing the issues we saw with flight testing. Additionally, as the glider never reached steady glide, our "true" glide angle and speed were more reflective of the start-to-end position of the glider test, not the actual glider's flight characteristics.

Watching the flight recordings, we saw that the glider appeared to temporarily fly at an angle around 5-10 degrees, which is much less than the 29-degree glide angle from the direct start-to-finish flight testing results. If the glider achieved a steady glide, we would most likely see a glide angle and speed more reflective of our calculated results.

If we had been able to realize this error sooner, we could have made several changes to the glider design and manufacturing to achieve steady glide. To do so, we would need to move the CG forward of the Neutral Point. The primary issue is that the glider is extremely tail heavy—the CG is almost 14" aft of the main wing's aerodynamic center. The following changes would contribute to achieving this goal.

1. Add weight to the front of the glider.
2. Shift the main wing rearwards on the fuselage.
3. Decrease the planform size of the main wing with respect to the tail.

4. Increase the aspect ratio of the main wing to reduce the downwash effect on the tail.
5. Reduce overall weight of the components.

Overall, the glider would require significant changes to be stable. For example, 19 ounces of weight would need to be added to the nose cone to achieve a MOS of 0. This is almost half the weight of the current glider.

Our team has gone through the process of designing, predicting the performance, manufacturing, and flight testing a glider of our own. While we would have liked to have better results, we now know where our design changes need to be made to achieve better results.